

1 Notation & Perturbation Method

Big $\mathcal{O}$: $f(z) = \mathcal{O}(g(z))$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} \text{ bdd} \Leftrightarrow |f(z)| \leq A|g(z)|$ as $z \rightarrow z_0$

Small o : $f(z) = o(g(z))$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = 0 \Leftrightarrow f(z) \ll g(z)$ as $z \rightarrow z_0$

Asymptotic $\sim$: $f(z) \sim g(z)$ as $z \rightarrow z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = 1$

Asymptotic Sequence: Seq $\{\varphi_n(z)\}$ is asy. if $\forall n \in \mathbb{N} \varphi_{n+1}(z) = o(\varphi_n(z))$ as $z \rightarrow z_0$

Asymptotic Expansion: $f(z) \sim \sum_{n=0}^M a_n \varphi_n(z)$ as $z \rightarrow z_0$ to M if $\phi_n(z)$ is asy. seq. and

$f(z) - \sum_{n=0}^N a_n \varphi_n(z) = o(\varphi_N(z)) = \mathcal{O}(\varphi_{N+1}(z)), \forall N \leq M$ as $z \rightarrow z_0$

$f(x, \varepsilon) \sim \sum_{n=0}^{\infty} a_n(x) \varepsilon^n$ uniformly in $x \in I$ as $\varepsilon \rightarrow 0$ if ε 与 x 无关 (i.e. $\sup_{x \in I} |f(x, \varepsilon) - \sum_{n=0}^N a_n(x) \varepsilon^n| = o(\varepsilon^N)$)

Singular|Regular - Perturbation Method: a. Let $\varepsilon = 0$ solve $f(x, 0) = 0$.

b.i. 如果 f 结构变了 (阶数降低), Let $x = \varepsilon^{-\alpha} X$, PDB 找 α c. 假设 $X = a_0 + a_1 \varepsilon^\beta + a_2 \varepsilon^{2\beta} + \dots$ 代入求解

b.ii. 如果 f 结构不变/变, 但有重根 x_0 , Let $x = x_0 + \varepsilon^\alpha X$, PDB 找 α . $\uparrow \beta$ 看化简后的 X 方程得出

Principle of dominant balance (PDB): For Singular Perturbation $p(x, \varepsilon) = 0$, Let $x = \varepsilon^\alpha X$, 代入选择 α :

对于 ε 来说 a. 至少有两项阶数相同且最低; b. 其他项阶数均更高. (即可忽略)

2 Integral By Parts & Laplace's Method

Taylor: $f(x) = \sum_{n=0}^{N-1} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n + \frac{f^{(N)}(\xi)}{N!} (x - x_0)^N \Rightarrow f(x) \sim \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$ as $x \rightarrow x_0$

Exp. Integral: $E_1(x) = \int_x^{\infty} \frac{e^{-t}}{t} dt$, ($x \rightarrow 0$ 发散) $E_1(x) \sim e^{-x} \sum_{n=0}^{\infty} (-1)^n \frac{n!}{x^{n+1}}$, $x \rightarrow \infty$

Gamma Func: $\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$ $\Gamma(x+1) = x\Gamma(x)$ and $\Gamma(n) = (n-1)!$ $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Incomplete Gamma Func: Lower: $\gamma(a, x) = \int_0^x t^{a-1} e^{-t} dt$ Upper: $\Gamma(a, x) = \int_x^{\infty} t^{a-1} e^{-t} dt$

$-\gamma(a, x) = \Gamma(a) - \Gamma(a, x)$ $\Gamma(a, x) \sim e^{-x} x^{a-1} \sum_{n=0}^{\infty} \frac{\Gamma(a)}{\Gamma(a-n)} x^{-n}$, $x \rightarrow \infty$

Laplace's Method: ????? If $I(x) = \int_a^b e^{-x f(t)} dt$ a. $f(t)$ has unique min at $t = t_0 \in (a, b)$

b. $f'(t_0) = 0, f''(t_0) > 0$ c. Taylor of $f(t) = f(t_0) + \frac{f''(t_0)}{2} (t - t_0)^2 + \dots$

Then, as $x \rightarrow \infty$, $I(x) \sim \int_{-\infty}^{\infty} e^{-x[f(t_0) + \frac{f''(t_0)}{2} (t-t_0)^2]} dt$

Remark: If $t_0 = g(x)$, Let $t = g(x)s$, then $I(x) = g(x) \int_{a/g(x)}^{b/g(x)} e^{-x f(g(x)s)} ds$ min at $s = 1$.

Application: $\int_{-\infty}^{\infty} e^{-x \cosh \theta} d\theta \sim e^{-x} \sqrt{\frac{2\pi}{x}}$ $\Gamma(n+1) = n! \sim \sqrt{2\pi n} (\frac{n}{e})^n$ (Stirling Formula)

3 Appendix

$y' + Py = Q: \mu y = \int Q(x) e^{\mu} dx, \mu = e^{\int P(x) dx} \downarrow x \geq 0$

Gaussian Integral: $\int_{-\infty}^{\infty} e^{-ax^2} dx = \frac{\sqrt{\pi}}{\sqrt{a}}$ **Inequalities:** $\ln(x+1) < x, e^x > x+1, (1+x)^a < e^{ax}$

Binomial Expansion: $(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots, |x| < 1$

Taylor: $f(x+h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!} h^n = f(x) + f'(x)h + \frac{f''(x)}{2!} h^2 + \dots$

$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \quad \cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \quad e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \quad \sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \quad \cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$

$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad \tan(x) = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots \quad \cot(x) = \frac{1}{x} - \frac{x}{3} - \frac{x^3}{45} - \dots$

Integral: $\int \tan x = -\ln |\cos x| = \ln |\sec x| \quad \int \cot x = \ln |\sin x| = -\ln |\csc x| \quad \int_0^{\infty} e^{-x} t^n dx = \frac{1}{n} \Gamma(\frac{n+1}{n}) x - \frac{m+1}{n}$